ITA0610-Machine Learning

LAB EXPERIMENTS

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Lab 28: NumPy Indexing and Filtering

Aim

To demonstrate **slicing, indexing, and boolean filtering** on multi-dimensional NumPy arrays.

Algorithm

1. Import the NumPy library.
2. Create a one-dimensional range of numbers and reshape it into a 2D array.
3. Access a specific element using indexing.
4. Access a complete column using slicing.
5. Apply boolean filtering to select elements greater than a specified value.
6. Use a boolean condition to modify selected elements.
7. Display the results.

Program

```
# Lab 28: NumPy Indexing and Filtering


import numpy as np


# Create a 2D array

a = np.arange(12).reshape(3, 4)


print("Array:\n", a)


# Access a specific element

print("\na[1, 2]:", a[1, 2])


# Access the second column

print("Column 1:", a[:, 1])
```

```
# Boolean filtering
print("Elements > 5:", a[a > 5])


# Conditional assignment
a[a % 2 == 0] = 0


print("\nAfter zeroing evens:\n", a)
```

Sample Output

```
Array:
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]]

a[1, 2]: 6
Column 1: [1 5 9]
Elements > 5: [ 6  7  8  9 10 11]

After zeroing evens:
[[ 0  1  0  3]
 [ 0  5  0  7]
 [ 0  9  0 11]]
```

Result

Thus, NumPy indexing, slicing, boolean filtering, and conditional assignment operations were successfully performed.